MILITARY SPECIFICATION

ALUMINUM CHLORIDE (ANHYDROUS)
(For Use in Ammunition)

This specification approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions and packaging for the manufacture of one grade of aluminum chloride (anhydrous) for use in thionyl chloride reserve type battery. (See 6.3)

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipping and Storage
MIL-STD-1218 - ACS Chemicals

FSC 6810

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM-E203 Water Using Karl Fischer Reagent
ASTM-E300 Sampling Industrial Chemicals
ASTM-E394 Iron in Trace Quantities Using the 1, 10-Phenanthroline Method
ASTM-D2905 Statement of Number of Specimens required to Determine the Average Quality of Textile Material.

(Application for copies should be addressed to the American Socieity for Testing and Materials, 1916 Race Street, Philadelphia, PA 19013).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The material shall be anhydrous aluminum chloride, a white crystalline solid having D values which conform to the D values as listed in the x-ray powder data files of ASTM when tested in accordance with 4.6.6 and shall conform to Table I.

TABLE I

Property	Requirements	Applicable Paragraph
Aluminum Chloride, Percent %, Min.	99.0	4.6.1
Water-insoluble Material %, Max.	0.3	4.6.2
Water %, Max.	0.8	4.6.3
Iron PPM, Max.	200	4.6.4
Free Aluminum PPM, Max.	100	4.6.5

3.2 Foreign matter. The aluminum chloride shall be free of dirt, chips and other foreign matter.

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3.3 First Article Inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of aluminum chloride obtained by sampling as described in 4.5.1 (smaller weight samples may be taken that are consistent with the requirements of the test). The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production (see 6.3).

4.3.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations for tests specified in 4.6 of this specification.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality Conformance Inspection.

4.4.1 Lot Formation. A lot shall consist of one or more batches of aluminum chloride produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of aluminum chloride that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous (See MIL-A-48078). The product shall be submitted for inspection in accordance with MIL-STD-105.

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4.5 Testing (see 6.2 and 6.4). PRECAUTION: This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Sampling. Two (2) individual 200 grams samples from each batch or lot of the aluminum chloride shall be sampled as specified in ASTM E300, for solids. The sample shall be tested independently in accordance with the applicable methods given in 4.6.

If either sample fails to meet any of the test requirements, the batch or lot shall be rejected. The samples shall be subjected to the tests listed below:

Inspection/Requirement	Defect Classification
Material (see 3.1)	Major
Properties, Table 1 (see 3.1)	Major

4.5.2 Inspection equipment. For the performance of all tests and examinations specified in 4.6, commercial inspection equipment should be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes.

4.6 Test methods and procedures (See 6.2 and 6.4). The tests to determine compliance with the requirements given in Table I shall be performed on the aluminum chloride samples obtained in 4.5.1. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals shall be reagent grade or ACS Grade in accordance with MIL-STD-1218.

4.6.1 Aluminum chloride.

4.6.1.1 Preparation of sample solution. Thoroughly clean a 300-ml extraction flask with a 24/40 outer joint neck, and dry the flask in an oven. Cool in a desiccator, close the flask with a No. 5 rubber stopper, and weigh to the nearest mg. Quickly add 4.0 to 4.2g of the sample, close the flask with the stopper, and reweigh to the nearest mg.

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Remove the stopper and insert a 20-in. West type condenser (24/40 inner joint) in the neck of the flask. Place a plug of glass wool, about two inches long, in the top of the condenser, so that the top of the plug is about two inches below the top of the condenser. Place the flask, supported loosely with a clamp on a ring stand, in an ice bath.

First wet the glass wool with distilled water avoiding more than a drop or two of water dropping into the flask. Now, add a small portion of distilled water through the condenser so that it enters the flask dropwise. When the water has reacted, raise the flask from the ice bath to warm the flask slightly. Replace the flask in the ice bath, add a little more water through the glass wool, and proceed in this manner until the sample is dissolved. Regulate the water addition carefully to avoid too vigorous evolution of hydrochloric acid from the sample. The alternate cooling and warming of the flask serves to draw the added water into the flask through the glass wool plug. When the sample is completely dissolved, rinse the glass wool and condenser with distilled water, catching the wash water in the flask.

Cool the solution and transfer to a 500-mL volumetric flask. Rinse the flask with water and add the rinsings to the volumetric flask. Dilute with water to the mark at 20°C, mixing the contents thoroughly.

4.6.1.2 Reagents. (See 4.6)

Aluminum chloride standard solution,
1 ml=0.01333 g $AlCl_3$
Barium hydroxide, saturated solution
Hydrochloric acid standard, 0.3 N
Potassium fluoride, 30% W/V
Sodium hydroxide, 1 N, approx.
Potassium sodium tartrate, 20% W/V

4.6.1.3 Procedure. With a pipet, transfer a 25.0 mL aliquot of the prepared sample solution, at 20°C, to a 600-mL beaker. Add 150 mL of 20% potassium sodium tartrate solution and, using a pH meter, adjust the pH to 6.5 with 1 N (approx.) sodium hydroxide. Titrate with saturated barium hydroxide solution to pH 8.3, add 5 mL excess of barium hydroxide solution and stir. Slowly add, dropwise, 0.3 N hydrochloric acid to pH 8.3. Add 30 mL of 30% potassium fluoride solution, previously adjusted to pH 8.3, and titrate the liberated potassium hydroxide slowly, dropwise, with 0.3N hydrochloric acid, until the solution remains at pH 8.3 for at least 30 seconds.

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4.6.1.4 Calculation.

$$\text{Percent Aluminum Chloride} = \frac{(A)(B)100}{W}$$

Where:
A = mL of Titrant
B = Grams of Aluminum Chloride per mL 0.3N Hydrochloric Acid
W = Grams of Sample Aliquot

4.6.2 Water insoluble material. (See 4.6.4)

4.6.2.1 Procedure. Weigh 30 to 40 g of the sample to the nearest 0.1g in a weighing bottle. Add the sample, about 5 g at a time, to a 600-mL beaker containing about 300 mL of distilled water. Keep the beaker covered with a watch glass and in a cooling bath, while the sample dissolves. After the added portion has been reacted, stir the solution until the temperature drops to about 30°C before adding more sample. Rinse out the weighing bottle and cover.

Filter the contents of the beaker through a tared medium Gooch which has previously been dried to constant weight at 110°C. Rinse the beaker with hot water to transfer all the insolubles to the crucible. Wash the crucibles several times with 10-ml portions of hot water. (Save the filtrate for the iron determination 4.6.4).

Dry the crucible containing the insoluble for two hours at 110°C and weigh.

4.6.2.2 Calculation.

$$\text{Percent Water Insoluble} = \frac{(A)100}{W}$$

Where:
A = Grams of Insoluble
W = Grams of Sample

4.6.3 Water. The water shall be determined in accordance with ASTM-E203.

4.6.3.1 Apparatus.

4.6.4 Iron The iron shall be determined in accordance with ASTM-E394 using the filtrate from 4.6.2.

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4.6.5 Free Aluminum.

4.6.5.1 Apparatus.

Crystallizing dish, Pyrex 190mm diameter
Magnetic stirrer
Thermometer, Laboratory, Range -10 or -20 to + 110°C. Subdivision 1°C
Thermometer clamp

4.6.5.2 Reagents. (see 4.6)

Water, distilled or deionized.

Sulfuric acid 1:3 Cautiously and with continuous stirring, add one volume of sulfuric acid, to 3 volumes of water.

Ferric chloride solution, 100 gpl.

Dissolve 100 grams of Ferric chloride hexahydrate in water, dilute to 1 liter and mix thoroughly.
Potassium dichromate solution standard 0.0100 N.
Weigh out exactly 0.4904 gram of potassium dichromate previously dried at 150° - 200°C.
Transfer to a 1 liter volumetric flask, dissolve in water and dilute to the mark with water at 20°C and mix thoroughly.

Phosphoric acid 85%

Sodium diphenylamin sulfonate indicator 0.25%.
Dissolve 125mg. of p-diphenylaminesulfonic acid sodium salt in water and dilute to 50 ml.

4.6.5.3 Procedure. Accurately pipet 50 ml of the ferric chloride solution into a 400 ml beaker containing 100 ml of water and a magnetic stirring bar. Place the beaker in an ice bath in a large crystallizing dish and set the dish on the top of a magnetic stirrer. Immersed a thermometer in the solution and cool the solution well below 20°C with the magnetic stirrer turned on.

Weigh 10.0 ± 0.1 grams of aluminum chloride sample in a dry 50 ml beaker. Slowly add the sample (a little at a time) to the cold ferric chloride solution, which is being continuously stirred. Add the sample slowly so that the temperature does

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not rise above 25°C. Remove the beaker from the ice bath after the sample has been completely dissolved. Add 4.5 ml of 1.3 sulfuric acid and mix. Then, add 4.5 to 5.0 ml of 85.% phosphoric acid and mix. Add 6 to 7 drops of 0.25% sodium diphenylamine sulfonate indicator and titrate with 0.0100N potassium dichromate solution to a violet end point.

Run a blank using the ingredients as above and titrate with 0.0100N potassium dichromate to a violet end point.

4.6.5.4 Calculations.

$$\text{ppm Free Al.} = \frac{(A-B)\ N\ (8.994)\ (1000)}{W}$$

Where:

A = mls of $K_2\ Cr_2O_7$ for sample.
B = mls of " " " for blank.
N = Normality of $K_2\ Cr_2O_7$.
W = grams of sample.

4.6.6 Material x-ray diffraction pattern. Determine the x-ray diffraction pattern of the sample using any standard powder diffraction method appearing in a standard of reference textbook (see 6.2).

4.6.7 Foreign matter. The aluminum chloride shall be examined visually for foreign material.

5. PACKAGING

5.1 Level C. The material shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules and Container Specifications for rail shipment or with National Motor Freight Rules and Container Specifications for truck shipments as applicable.

5.2 Marking. Unless otherwise specified, the shipping containers shall be marked in accordance with MIL-STD-129, and shall include the lot number.

6. NOTES

6.1 Ordering data. See MIL-A-48078

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6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.3 Aluminum chloride which has been found satisfactory for use in lithium batteries is manufactured by Fluka AG Buchs Anal. ABT, CH-9470 Buchs, Switzerland. Aluminum chloride from other sources should be evaluated by ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801 before being approved as an equivalent.

6.4 The tests described in this specification should be performed in an inert atmosphere glove box, or in a dry room with relative humidity not to exceed 5% at ambient conditions.

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nger of the primary samples shall be identification numbers. The remai

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